

**JAMES RUSE AGRICULTURAL HIGH SCHOOL**  
**1998 4 UNIT MATHEMATICS**  
**TERM 4 ASSESSMENT**

Time: 85 minutes.

No equipment may be borrowed during the examination.

All necessary working must be shown.

**Question 1**

- (a) If  $z = 2 - 3i$  and  $\omega = -1 - i$ , find
- (i)  $|z|$
  - (ii)  $\arg \omega$
  - (iii)  $\omega \bar{z}$
- (b) Find  $\sqrt{-5 - 12i}$  in the form  $a + ib$ .
- (c) If  $z = 1$  is one of the roots of the equation  $iz^2 + (ia - 1)z + (i - b) = 0$ , where  $a$  and  $b$  are real, find:
- (i) the values of  $a$  and  $b$ .
  - (ii) the other root.
- (d) If  $z = 4\lambda + 3i(1 - \lambda)$  and  $\lambda$  is real,
- (i) prove that the point  $P$  which represents  $z$  on the Argand plane lies on a straight line.
  - (ii) find the least value of  $|z|$ .

**Question 2 ( Start a new page )**

- (a) Express in mod-arg form:
- (i)  $\frac{5 + i}{2 + 3i}$
  - (ii)  $i\bar{z}$  if  $z = 4(\cos \theta + i \sin \theta)$ .
- (b) Find all  $n$  for which  $(2 - 2\sqrt{3}i)^n$  is real.
- (c) What single transformation maps  $z$  to:
- (i)  $\bar{z}$
  - (ii)  $iz$
- (d) Show that there is only one value of  $z$  for which both  $\arg(z + i) = \frac{\pi}{4}$  and  $|z + 1| = 2$ , and that this value is real.

### Question 3 (Start a new page )

- (a) Given  $z = x + iy$  and  $P(x,y)$  moves in the complex plane, graph the locus of  $P$  and state its Cartesian equation if:
- $|z - i| = |z + 1|$
  - $\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{2}$
  - $\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$
- (b) Using suitable diagram(s) on the Argand plane, prove the following inequalities from **geometric considerations** alone. Give clear reasons.
- $|z_1 + z_2| \leq |z_1| + |z_2|$ .
  - $|z - 1| \leq ||z| - 1| + |z| |\arg z|$ , where  $-\pi < \arg z \leq \pi$ .

### Question 4 (Start a new page )

- (a) If  $\alpha = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ ,  $\beta = \alpha + \alpha^2 + \alpha^4$  and  $\gamma = \alpha^3 + \alpha^5 + \alpha^6$ ,
- prove that  $\beta + \gamma = -1$  and  $\beta\gamma = 2$ .
  - Hence show that:  $\sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7} = \frac{\sqrt{7}}{2}$ .
- (b) In the Argand plane, consider the triangle  $ABC$  with vertices at  $A(-1,0)$ ,  $B(1,0)$  and  $C$  represents the complex number  $z$ , for  $-\pi < \arg z \leq \pi$ .  
Let  $L$  be a straight line through the origin and parallel to the bisector of  $\angle ACB$ .  
Prove that both values of  $\sqrt{z^2 - 1}$  lie on the line  $L$ , clearly labelling all important features in your diagram.
- (c) The points  $P_1, P_2, P_3, \dots, P_n$  representing the complex numbers  $z_1, z_2, z_3, \dots, z_n$  lie below the straight line  $y = \sqrt{3}x$ .
- The points  $Q_1, Q_2, Q_3, \dots, Q_n$  representing  $\frac{1}{z_1}, \frac{1}{z_2}, \frac{1}{z_3}, \dots, \frac{1}{z_n}$  possess a similar property. State this property and the equation of the corresponding straight line.
  - Show that  $z_1 + z_2 + \dots + z_n \neq 0$ , for **any** set of  $n$  points, and similarly  $\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \dots + \frac{1}{z_n} \neq 0$ .

**END of PAPER**

## QUESTION 1.

[4]

a) (i)  $|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$  (1)

(ii)  $\arg w = -\frac{3\pi}{4}$  (1)

(iii)  $w\bar{z} = (-1-i)(2+3i)$   
 $= -2-3i-2i+3$   
 $= 1-5i$  (2)

[3]

b)  $\sqrt{-5-12i} = x+iy$   
 $-5-12i = (x+iy)^2$   
 $\begin{cases} x^2 - y^2 = -5 \\ 2xy = -12 \end{cases} \Rightarrow y = \frac{-6}{x}$  (3)

$\therefore x^2 - \frac{36}{x^2} = -5$   
 $x^4 + 5x^2 - 36 = 0$   
 $\therefore x = 2 \text{ or } x = -2$   
 $y = -3 \text{ or } y = 3$   
 $\therefore \text{roots } 2-3i, -2+3i$  (1)

[5]

c) (i)  $i + ia - 1 + i - b = 0$   
 $\begin{cases} -1-b = 0 \\ 2+a = 0 \end{cases} \therefore b = -1, a = -2$  (3)

(ii)  $\alpha + \beta = -a + \frac{1}{i} = 2 - i$  (2)  
 $\therefore \alpha = 1$   $\therefore$  other root  $1-i$

[3]

d) (i)  $z = 4\lambda + 3i(1-\lambda)$  where  $z = x+iy$   
 $\begin{cases} x = 4\lambda \\ y = 3(1-\lambda) \end{cases} \Rightarrow \lambda = \frac{x}{4}$   
 $y = 3 - \frac{3x}{4}$  (2)

$\therefore P$  lies on the straight line  $3x+4y=12$

(ii)  $d = \frac{|10+0-12|}{\sqrt{3^2+4^2}} = \frac{2}{5}$  (1)

## QUESTION 2.

[6]

a) (i)  $\frac{5+i}{2+3i} = \frac{(5+i)(2-3i)}{(2+3i)(2-3i)} = \frac{13-13i}{13} = 1-i$  (1)

(ii)  $i\bar{z} = i(4\cos\theta - 4i\sin\theta) = 4(\sin\theta + i\cos\theta)$  (3)  
 $= 4\left(\cos\left(\frac{\pi}{2} + \theta\right) + i\sin\left(\frac{\pi}{2} + \theta\right)\right)$

[5]

b)  $(2-2\sqrt{3}i)^n = \left[4\left(\cos\frac{-\pi}{3} + i\sin\frac{-\pi}{3}\right)\right]^n$   
 $= 4^n \left(\cos\frac{-n\pi}{3} + i\sin\frac{-n\pi}{3}\right)$  (3)

Real for  $\sin\frac{-n\pi}{3} = 0$  (2)  
 $\therefore \frac{n\pi}{3} = k\pi \Rightarrow n = 3k, k \text{ integer}$

[2]

c)  $z \rightarrow \bar{z}$  reflection on X axis (1)  
 $iz \rightarrow i\bar{z}$  rotation of  $\frac{\pi}{2}$ , anticlockwise (2)

d)

$\begin{cases} \arg(z+i) = \frac{\pi}{4} \\ |z+i| = 2 \end{cases} \Rightarrow \begin{cases} y = x-1, x > 0 \\ (x+1)^2 + y^2 = 4 \end{cases}$  (1)

$\therefore 2x^2 + 2 = 4 \Rightarrow x = \pm 1$  (4)  
 but  $x > 0 \therefore x = 1$  is the only solution  
 $\therefore z = 1$

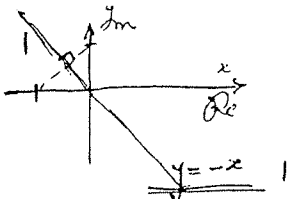
[4]

if from  
diagram  
give 4 marks

### QUESTION 3.

a)

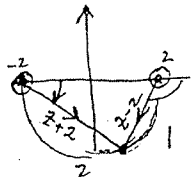
(i)



$$|z-i| = |z+i|$$

(2)

(ii)



$$\arg\left(\frac{z+2}{z-2}\right) = \frac{\pi}{2}$$

$$x^2 + y^2 = 4, y < 0$$

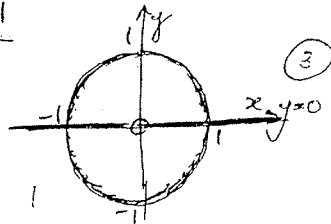
(3)

(iii)

$$\operatorname{Im}\left(z + \frac{1}{z}\right) = 0$$

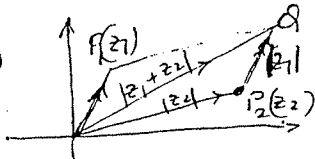
Either  $z$  real  $\neq 0$  or  $\frac{1}{z} = -\bar{z}$

$$\therefore y = 0, x \neq 0, \quad x^2 + y^2 = 1$$



(3)

b) (i)



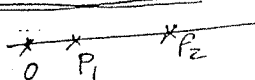
$OP_1, OP_2$  given by construction

In  $\triangle OP_1P_2$ ,  $OP_1 < OP_2 + P_1P_2$

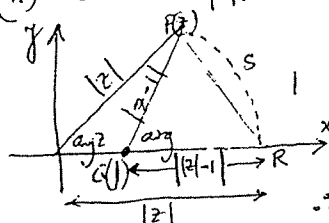
"each side of a  $\triangle$  is less than the sum of the other two sides"

$$\therefore |z_1 + z_2| < |z_1| + |z_2|$$

But if  $OP_1, P_2$  are in a straight line then the equality stands  $|z_1 + z_2| = |z_1| + |z_2|$



(ii) Case 1.  $|z| > 1$



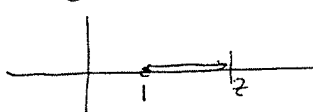
$$|z-1| < |z|-1 + |PR|$$

but arc  $BR = |z| \arg z$   
 $l = r\theta$ ,  $|PR| < \text{arc}$

$$\therefore |z-1| < |z|-1 + |z| \arg z$$

(4)

Case 2.  $z$  real with  $\theta = 0$



$$\therefore |z-1| = ||z|-1|$$

$\therefore$  equality applies since  $|\arg z| = 0$

$$|z-1| = ||z|-1| + |z| |\arg z|$$

Bonus mark if student considers  $|z| < 1$  and  $|z| = 1$  chord less

### QUESTION 4.

a)

$$z^7 = 1 = \cos 2\pi n + i \sin 2\pi n \Rightarrow x = \cos \frac{2\pi n}{7} + i \sin \frac{2\pi n}{7}$$

$$\rho = \begin{pmatrix} \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \\ \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \\ \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7} \end{pmatrix} = \begin{pmatrix} \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7} \\ \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \\ \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \end{pmatrix} \quad \alpha^7 = 1$$

$$\beta + \gamma = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 + (\alpha^7 - \alpha^7) = (\text{sum of roots of } z^7 - 1 = 0) - 1 = 0 - 1 = -1$$

(2)

[7]

$$\beta \gamma = (\alpha + \alpha^2 + \alpha^3)(\alpha^3 + \alpha^5 + \alpha^6) = \alpha^4 + \alpha^6 + 1 + \alpha^5 + 1 + \alpha + 1 + \alpha^2 + \alpha^3 = \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6 + 3 = -1 + 3 = 2$$

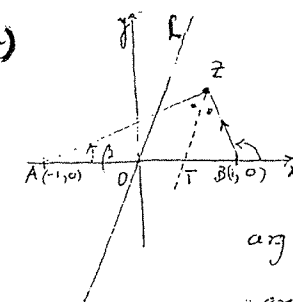
$$(ii) \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{6\pi}{7} = \operatorname{Im}(\beta)$$

$$\text{To find } \beta \text{ of } x^2 + x + 2 = 0 \therefore x = \frac{-1 \pm \sqrt{7}i}{2}$$

$$\therefore \beta = \frac{-1 + \sqrt{7}i}{2}, \quad +\sqrt{7} \text{ since } \sin \frac{4\pi}{7} > \sin \frac{6\pi}{7}$$

$$\therefore \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{6\pi}{7} = \frac{\sqrt{7}}{2}$$

b)



$$\arg(z-1) = \alpha, \quad \arg(z+1) = \beta$$

$$\angle ZAB = \frac{\alpha - \beta}{2} \therefore \angle ZTB = \frac{\alpha + \beta}{2}$$

(external angle of  $\triangle ZAB$ ) (angle sum of  $\triangle ZTB$ )

$$\therefore \angle LOX = \frac{\alpha + \beta}{2}$$

$$\arg \sqrt{z^2 - 1} = \frac{1}{2} [\arg(z-1) + \arg(z+1)]$$

$$\therefore \arg(\sqrt{z^2 - 1}) = \frac{1}{2} (\alpha + \beta) = \angle LOX$$

$\therefore$  both values of  $\sqrt{z^2 - 1}$  lie on line L.

c) (i) Since  $\arg \frac{1}{z} = -\arg z$  the points representing  $\frac{1}{z_1}, \frac{1}{z_2}, \dots, \frac{1}{z_n}$  lie above the line  $y = -\sqrt{3}x$

(ii) By rotating the full plane  $\frac{\pi}{6}$  anticlockwise, all points will have  $x > 0$   $\therefore$  let  $w = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

$$\therefore w z_1 + w z_2 + \dots + w z_n \neq 0 \quad \text{since } \operatorname{Re}(w z_i) > 0$$

$$\therefore \operatorname{Re}(w z_1 + w z_2 + \dots + w z_n) > 0$$

$$\therefore w(z_1 + z_2 + \dots + z_n) \neq 0 \quad \text{since } z = 0 \text{ iff } x = 0$$

$$\therefore z_1 + z_2 + \dots + z_n \neq 0$$

Similarly for  $L \perp \dots \perp$  with  $w = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$